PMCO-AI: Personalized Motivation, Capability, and Opportunity in the AI Era — A Case Illustration in Dementia Care

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Abstract

Traditional public health and healthcare behavior-change approaches often emphasize personal responsibility, relying on motivation or static instructions. While widely applied, these approaches frequently fail in complex human systems, particularly where cognitive or environmental constraints exist. The PMCO-AI framework (Personalized Motivation, Capability, and Opportunity powered by AI) extends the COM-B model by integrating AI-driven personalization, real-time pattern recognition, and context-sensitive execution. This paper illustrates PMCO-AI using dementia care as a case scenario, emphasizing opportunity-first thinking. The framework demonstrates how AI can adapt to unfolding reality, create actionable options, and enhance human judgment without replacing responsibility. Implications extend to chronic disease management, aging services, and public health interventions in contexts characterized by complexity, uncertainty, and distributed responsibility.

Keywords: *PMCO-AI, Personalized Behavior Change, Opportunity-First, Dementia Care, AI in Health, Human-AI Partnership, Adaptive Interventions, Real-Time Decision Support*

1. Introduction

Public health has historically relied on moralized notions of personal responsibility or generalized encouragement for behavior change. Messages such as "eat healthier," "exercise more," or "get vaccinated" implicitly place the burden on individuals, often without providing the skills, resources, or context needed for effective action. Evidence indicates that blame rarely motivates lasting change, and encouragement without capacity-building is insufficient, particularly when structural barriers exist, such as poverty, limited access, or competing responsibilities.

The COM-B model (Capability, Opportunity, Motivation → Behavior) provides a useful framework for designing behavior-change interventions[1]. However, even COM-B has limitations in complex, dynamic systems, where behavior depends on rapidly changing contexts and fluctuating human capacities.

To address these limitations, this paper applies the PMCO-AI framework (Personalized Motivation, Capability, and Opportunity powered by AI), a framework proposed by the author in a Springer textbook on AI in public health[2]. PMCO-AI extends COM-B by embedding AI-driven personalization into each component, enabling real-time, context-sensitive behavior-change interventions. AI supports continuous monitoring, pattern recognition, and dynamic adaptation, allowing interventions to respond effectively to environmental, cognitive, and social constraints.

Dementia care illustrates these challenges vividly. Care plans may appear well-structured in design, yet real-world execution often diverges due to unpredictable patient behavior, caregiver constraints, and fluctuating cognition. This paper demonstrates how PMCO-AI shifts intelligence from static design to opportunity-focused execution, providing actionable guidance that complements human judgment and improves care outcomes.

2. Conceptual Frameworks

Behavior-change frameworks in public health have evolved from purely motivational approaches to integrated models such as COM-B. COM-B emphasizes that behavior results from interactions among capability, opportunity, and motivation. Despite its utility, COM-B does not fully address the challenges of real-time adaptation in complex human systems or leverage AI for personalization.

PMCO-AI integrates AI as a core enabler across all behavioral components. AI supports real-time sensing, pattern recognition, and dynamic adaptation, allowing interventions to respond to context and human capacity constraints. The framework shifts from planning-focused logic to execution-layer intelligence, enabling practical, context-sensitive guidance in complex systems such as dementia care.

3. PMCO-AI Framework

3.1 Definition and Scope

PMCO-AI combines three behavioral dimensions with AI-driven personalization (Figure 1), which orders behavioral considerations rather than operational logic. PMCO-AI is a conceptual starting point—local deployments must add triage, bias audits, and equity-aware allocation.

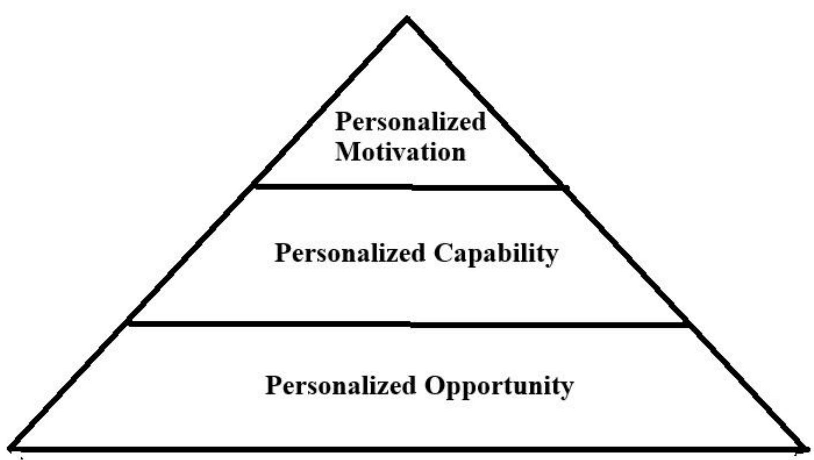

Figure 1. PMCO-AI conceptual framework

1. **Personalized Motivation (PM):** Tailored encouragement aligned with intrinsic drivers, social norms, or situational context.
2. **Personalized Capability (PC):** Support for skills, knowledge, and practical tasks adapted to individual strengths and cognitive levels.
3. **Personalized Opportunity (PO):** Context-sensitive, actionable options that are feasible, timely, and aligned with real-world constraints.

AI serves as the enabling mechanism, adapting interventions dynamically. Unlike traditional AI systems focused solely on prediction, PMCO-AI emphasizes real-time execution, helping humans navigate uncertainty and maximize impact.

3.2 Design Principles

Key PMCO-AI principles include:

- Opportunity-first prioritization: Feasible actions are essential; without actionable opportunities, motivation or capability alone cannot produce behavior change.
- Context-sensitive adaptation: Real-time sensing allows AI to adjust recommendations to environmental, temporal, or social constraints.
- Human-AI partnership: AI supports but does not replace human judgment. Responsibility remains with caregivers or individuals.
- Scalability and personalization: AI enables individualized interventions at scale, making complex behavioral guidance practical for public health.

4. Implementation and Operational Logic

PMCO-AI operationalizes the framework through execution-layer AI systems that:

1. Monitor environmental and individual context in real time.
2. Identify feasible interventions that are immediately actionable.
3. Support human decision-making with tailored prompts, reminders, and guidance.
4. Continuously update recommendations based on observed outcomes and changing conditions.

In practice, Opportunity serves as the primary enabler, particularly when cognitive or environmental constraints limit the efficacy of motivation- or capability-centered interventions. Motivation and capability remain relevant, primarily to guide caregiver engagement and ensure interventions are appropriate.

5. Illustrative Case: Dementia Care

Dementia care exemplifies the need for opportunity-first thinking. Patients' cognitive limitations and fluctuating behavior make strictly prescriptive care plans insufficient. Pre-defined predictive alerts may highlight risks but cannot generate immediately actionable options for caregivers. PMCO-AI addresses this gap by prioritizing opportunity, complemented by motivation and capability.

5.1 Opportunity-First Logic

Opportunity represents the feasibility of action within current constraints. In dementia care:

- Highly motivated caregivers cannot act effectively without feasible options.
- Patients with cognitive decline may have limited ability to respond to motivation or capability interventions.
- AI systems that focus on opportunity enhance caregiver effectiveness, reduce decision paralysis, and allow dynamic adaptation to changing conditions.

5.2 PMCO-AI Dimensions in Dementia Care

The illustrative cases for dementia care are shown in Table 1 and an illustrative case vignette for dementia care is attached to Appendix A below.

PMCO Dimension	Conceptual Role	Constraints in Dementia Care	Example AI-Guided Opportunities
Motivation	Tailored encouragement and affective guidance	Cognitive decline limits patient-initiated behavior; emotional responses fluctuate	AI provides caregiver-focused prompts to engage patients in feasible, enjoyable activities; reassurance cues
Capability	Practical skill support and guidance	Patients have limited independent decision-making; caregivers have variable expertise	AI delivers simplified instructions, visual cues, or reminders for caregivers to facilitate safe, achievable interventions
Opportunity	Context-sensitive, feasible actions	Environmental conditions, caregiver availability, and patient condition constrain options	AI continuously perceives context and recommends low-burden interventions, environmental adjustments, or safe activities; prioritizes immediately actionable options
Execution Principle	Shifts intelligence from design-layer plans to real-time action	Pre-defined rules or predictive alerts alone cannot ensure implementation	AI highlights feasible options and supports caregiver decision-making while preserving human judgment

Table 1. Illustrative cases for dementia care

This framework illustrates that opportunity-first execution is critical, especially when real-world constraints limit feasible actions. Motivation and capability remain secondary but help ensure interventions are appropriate and aligned with human judgment.

6. Safety, Trust, and Governance

AI-enabled behavior-change systems must address:

- Safety: Interventions must be low-risk, contextually appropriate, and regularly validated.
- Trust: Transparency in AI recommendations, explainability, and human oversight are essential for caregiver and patient adoption.
- Governance: Policies should define responsibilities, data usage standards, and accountability for AI-assisted decisions.

In dementia care, ensuring safe and trustworthy AI guidance reduces caregiver burden while maintaining patient well-being and compliance with ethical standards.

7. Implications

The PMCO-AI framework has broader relevance:

- Chronic disease management: Complex, multi-factorial conditions benefit from context-aware, opportunity-focused AI guidance.
- Aging services: Older populations with cognitive or physical limitations require actionable opportunities, not just alerts.
- Public health crises: Climate-related events or pandemics create dynamic environments where feasibility and rapid adaptation are crucial.
- Human-AI collaboration: AI enhances actionable options while preserving accountability and human judgment.

8. Conclusions

In complex human systems, including dementia care, traditional behavior-change approaches or predictive AI alone are insufficient. PMCO-AI demonstrates that:

- Opportunity-first execution is essential for translating plans into effective action.
- Capability and motivation remain relevant but often secondary to feasible action opportunities.
- AI-human collaboration enhances safety, reduces caregiver stress, and ensures context-sensitive, adaptive interventions.

The PMCO-AI framework provides a conceptual and practical guide for designing AI-enabled behavior-change systems in public health, aging services, and other domains characterized by complexity, uncertainty, and distributed responsibility.

Reference

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2. Wu M. *Artificial Intelligence in Public Health.* New York, NY, USA: Springer Nature; 2026.

Declarations

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Ethics Approval: This is a conceptual/perspective paper using an illustrative, non-identifiable case vignette and does not involve human subjects research or identifiable patient data.

Data Availability: No new data was generated or analyzed in this conceptual paper.

Appendix A: Illustrative Case Vignette – Dementia Care

A care plan for an individual with dementia may appear well-structured: medications are scheduled, meals are prepared, and caregiver check-ins are planned. Predictive AI alerts highlight potential risks such as missed medications, falls, or cognitive lapses.

In practice, execution rarely follows the plan perfectly. The individual may refuse meals, wander unexpectedly, or become disengaged from reminders. Caregivers often face competing demands and limited capacity, making rigid adherence to pre-defined schedules challenging. Standard AI alerts, while technically correct, provide little guidance for action when circumstances diverge from the plan.

A PMCO-AI approach addresses these challenges by shifting focus from prediction to opportunity-first execution, continuously perceiving the environment, the individual's state, and caregiver availability. The system generates context-sensitive options, helping caregivers act effectively in real time. For example:

- Opportunity: The AI detects moments when the individual appears calm and suggests a short, supervised walk or a simple, preferred snack, instead of insisting on the planned meal or activity.
- Capability: The AI provides caregivers with simplified instructions or visual cues, such as step-by-step reminders for administering medications safely, or guidance on assisting with hygiene and mobility.

- Motivation: The AI offers caregiver-focused prompts to encourage engagement, like suggesting a brief interaction that aligns with the individual's current mood or interests, reinforcing positive routines.
- Execution: The AI continuously adapts recommendations as circumstances change, for instance alerting caregivers to remove minor environmental hazards, adjust seating arrangements, or redirect wandering behavior, ensuring interventions remain feasible and low burden.

By presenting feasible options rather than rigid rules, PMCO-AI preserves human judgment and responsibility while expanding the caregiver's effective range of action. Capability and motivation remain relevant but are secondary to identifying immediate, actionable opportunities.

This opportunity-first, context-sensitive approach ensures that interventions remain practical even in dynamic, unpredictable scenarios. Caregivers can select and prioritize the most appropriate actions at each moment, maintaining the individual's safety, comfort, and engagement. The framework demonstrates how AI can complement human decision-making, reduce caregiver stress, and support adaptive, resilient care in complex systems like dementia management.